MICHAEL HE

MACHINE LEARNING ENGINEER

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Ottawa, ON

PROFESSIONAL SUMMARY

Machine learning engineer with a Master of Applied Data Science, 2 years of machine learning research experience, and 12 months of software engineering experience at Ciena. Experience includes PyTorch model experimentation, LLM data preparation, multi-agent AI systems, semantic retrieval, and Python application development.

WORK EXPERIENCE

Software Engineer and Tools Developer Co-op September 2025 - August 2026
Ciena *Ottawa, Ontario, Canada*

- Designed the orchestration workflow and developed five specialized agents for a deployed 10-agent AI debugging system, supporting automated root-cause analysis across four products.
- Improved data quality for domain-specific LLM training by filtering low-value content from 10,000 Confluence pages and generating descriptions for 3,000 embedded images using Qwen3-6B.
- Reduced loading time for 2 GB+ state-dump files from 1-2 minutes to approximately 20 seconds and resolved browser crashes through backend optimization.
- Developed REST APIs using FastAPI and Flask to integrate six AI agents with React interfaces, managing backend services and handling agent responses.

Graduate Student Researcher September 2021 - September 2023
Chongqing University *Chongqing, China*

- Improved cross-domain object detection mAP by reducing feature distribution differences in a PyTorch Faster R-CNN model across Cityscapes, Foggy Cityscapes, and SIM10K datasets.
- Built, trained, and evaluated a PyTorch Transformer to predict keywords in future book titles using a dataset of 20,000 historical titles.

Data Analyst Intern December 2020 - February 2021
Tianbojincheng Intelligent Medical Technology (Nanjing) *Nanjing, China*

- Built and trained a PyTorch deep learning model to fill missing values in a dataset containing 20,000 gene expression features across 40 tissue regions, supporting downstream data analysis.

EDUCATION

Master of Applied Data Science September 2024 - August 2026
University of Victoria
GPA: 8.21/9.0. Relevant coursework: Machine Learning, Optimization for Machine Learning, Pattern Recognition.

Bachelor of Computer Science and Technology September 2017 - June 2021
Hohai University
Relevant coursework: Pattern Recognition, Database Principle.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, SQL, R
- **Machine Learning & Computer Vision:** PyTorch, scikit-learn, Faster R-CNN, domain adaptation, Transformer models, model evaluation
- **Data:** Data preprocessing, dataset curation, metadata filtering, Milvus
- **Software Engineering:** FastAPI, Flask, React, REST APIs, Git, Docker, Gunicorn, Linux deployment

AWARDS & HONORS

- **Runner-up, Ciena AI Hackathon** Five-person team selected from more than 200 teams 2026
- **First-Class Scholarship (Category A), Chongqing University; Academic Excellence Scholarship, Hohai University**